ATLAS Independent Logical Audit

Basin Escape, Player-0 Activation, and Settlement Preorders

ATLAS-style review, prepared by ChatGPT

21 October 2026

Decision and scope

Verdict: CONDITIONAL PASS as a working research note; NOT YET PASS for an unqualified formally audited theorem paper. The central distinctions—actual first-person success versus possible success; adversarial chess versus static formal proof; verifier-gated lemma sharing; and reachable proofs versus measured speedup—are mostly sound and carefully qualified. A genuine gap occurs in the stated settlement-preorder theorem. A second gap occurs in the BANK lemma unless import eligibility is made an explicit assumption.

Audited artifact: *Basin Escape, Player-0 Activation, and Settlement Preorders*, 9-page PDF dated 16 October 2026, uploaded as Basin_Escape_Note(3).pdf. The findings below concern only the supplied manuscript. This is a **manual mathematical and protocol audit**: no external theorem prover was run, no computational experiment was initiated, and the four works in its bibliography were not independently checked against their full texts. The source URL for [1] could not be confirmed from the available public web access. Thus source fidelity and independent machine verification remain **NOT VERIFIED**, not failed.

Disposition matrix

ID	Severity	Location	Result
A01	**Blocking**	§8, Definition 8.1 / Theorem 8.2	Actual emission does not follow from simulability; membership on equivalence classes is underdefined.
A02	**High**	§6, Lemma 6.2	Dependency retention alone does not enforce sound downstream BANK imports.
A03	**High**	§8, bounded settlement	Resource transforms do not automatically preserve the same fixed budget.
A04	Moderate	§§3–5, first-person intervention A	The subject, handoff, and distinct operational effect of prompting must be specified.

ID	Severity	Location	Result
A05	Moderate	§8, “SIC”	Mutual simulation is not generally simulation *isomorphism*.
A06	Moderate	§§9–10, experiment design	Expected cost and failure/censoring treatment are not defined enough for comparisons.
A07	Advisory	§8, Theorem 8.4	Well-foundedness gives minimal antichains, but not necessarily finite ones.
A08	Advisory	§7, Proposition 7.1	Classicality and consistency are stronger assumptions than needed for usual negation.
A09	Advisory	§§4, 7, 10, provenance	Freeze C/K/U role status, checker interfaces, and source-verification limitations.

Mathematical findings and repairs

A01 — Blocking: actual-emission versus possible-execution semantics

Textual conflict. Definition 8.1 (p. 6) places $[X]$ in U_c when **X emits** an accepted certificate of $T \vdash c$ or $T \vdash \neg c$. However, the preorder $X \preccurlyeq Y$ is formulated through Y’s ability to **reproduce** relevant certified behavior, and Theorem 8.2 assumes that an accepted certificate produced by X **can be** reproduced or transferred to Y. Ability to produce is not identical to actual production. The proof silently replaces *may emit* with *does emit*.

Counterexample. Let X have one admissible successful run that outputs a valid proof in one step. Let Y include every legal step of X, but also an idle branch selected by Y’s frozen scheduler, which consumes the budget without output. Y can legally simulate X’s successful trace, yet its actual execution does not emit the proof. Thus $X \preccurlyeq Y$, $X \in U_c$, and $Y \notin U_c$ are simultaneously possible. Even worse, if mutual simulations preserve only available traces, a system’s actual success may not be invariant under $X \sim Y$; then membership of $[X]$ itself is **not well-defined**.

Required repair (choose one and apply consistently):

1. **Capability semantics (recommended for SIC ordering):** define $U^{\exists}_{c,b}$ by the **existence of a legal, resource-feasible** trace that emits an accepted settlement certificate. Define $\preccurlyeq_b$ to preserve every accepted successful trace, including the budget, and quotient by the corresponding symmetric relation. Under these hypotheses, the revised upward-closure theorem is correct. Actual experiments belong to a separate outcome relation.
2. **Actual-run semantics (recommended for first-person predictions):** include the complete machine identity, scheduler, seed, verifier, budget, activation history, and trace-selection convention. Demand that the simulation relation preserve the **actual certified output trace**, not merely its availability. This stronger relation yields an upper-set result for actual outputs, but not from ordinary capability simulation alone.

Minimal replacement theorem: Fix a common object task $q = (T,c,V)$, a bound b, and an admissible-run semantics. Let $S_b(X,c)$ mean *some legal X-run of charged cost at most b emits a*

V-accepted certificate for c or ¬c. Suppose $X \preccurlyeq_b Y$ guarantees that every such X-run maps to a legal Y-run whose charged cost is also at most b and whose certificate remains accepted. Then $U^{\exists}_{c,b} = \{[X] : S_b(X,c)\}$ is well-defined on mutual-$\preccurlyeq_b$ classes and is upward closed. The proof is immediate by applying the simulation to the successful witness run.

Propagation: Definitions 8.3 and 8.6 and Proposition 8.7 should refer to the repaired settlement region, or be explicitly restricted to a domain on which the region is well-defined.

A02 — High: BANK provenance is necessary, not an import rule

Lemma 6.2 (p. 5) assumes sound checking and preservation of a certificate's undischarged hypothesis set Γ. But the proof adds another, unstated requirement: "any later use must preserve its dependencies or provide" a valid discharge. That **consumer-side admissibility rule** is essential and should be part of the formal statement, not merely the proof.

Counterexample. In an empty base theory, the conditional sequent $p \vdash p$ has a perfectly sound trivial certificate. A BANK that retains $\Gamma = \{p\}$ has not laundered anything *yet*. If a receiving prover silently treats the imported p as unconditional, it can incorrectly claim $\vdash p$, although Γ was accurately recorded. Faithful storage does not prevent unsound consumption without an enforced import check.

Repair: Store $e = (T, \ell, \pi, \Gamma, \text{origin, cost, verifier-id, provenance})$ with $V(\pi, T \cup \Gamma \vdash \ell) = 1$. For importing into current context $T \cup \Delta$, require either (i) that every premise in Γ is already admitted in Δ, or (ii) independently checked derivations of $T \cup \Delta \vdash \gamma$ for **each** $\gamma \in \Gamma$. Composition/import itself must be verified or be a sound kernel rule. Reject all other promotions. Restate Lemma 6.2 with this rule among its explicit hypotheses.

A03 — High: simulation overhead and the fixed budget

Section 8 allows a "declared resource-accounting transformation," but that is not enough to preserve membership in a **same-budget** settlement region. An X trace with cost 1 may map to a Y trace costing 11; with bound $b = 1$, Y does not settle in that budget even though it correctly simulates X. A simulation cost map g generally yields $U_{c,b}(X) \Rightarrow U_{c,g(b)}(Y)$, not $U_{c,b}(Y)$.

Repair: Either index the preorder by budget and permit only mappings whose transformed charge does not exceed b, or state a cross-budget monotonicity theorem with the explicit map g. State how verifier calls, activation, routing, ledger operations, certificate transport, and parallel compute are charged. Preserve wall-clock time as a **distinct** outcome.

A04 — Moderate: first-person and federation semantics

Definition 3.1 (p. 3) correctly identifies the designated M and its actual output event, and correctly warns that a graduate federation's proof does not automatically satisfy a claim made about its parent. Definition 5.1 (p. 4) should carry this discipline into the activation rule: who speaks, which executable is frozen, which component's output counts, and whether a verified delegated result becomes the parent's emitted output must be specified **before the run**. An assertion that "I will prove c" is **not itself** a verifier-accepted proof of c.

A second issue is empirical identifiability: "A: self-settlement prompt" is a framing or task-assignment rule, not yet a distinct executable search algorithm. Specify the actual action-selection/update behavior induced by A, otherwise no causal basin-escape benefit can be assigned to that intervention rather than to other solver changes.

A05 — Moderate: SIC terminology

The §8 quotient $X \sim Y$ iff $X \preccurlyeq Y$ and $Y \preccurlyeq X$ is a legitimate quotient of a preorder and produces a partial order. But mutual simulation need not imply a structure-preserving isomorphism of the underlying simulations. If "SIC" already has a stricter meaning in the larger framework, this paper must

either prove coincidence in its intended model class, explicitly redefine the term, or use **mutual-simulation equivalence classes** here instead.

A06 — Moderate: performance outcomes and experimental comparison

Research Conjecture 9.2 (p. 8) is correctly labeled unproven. However, the proposed "lower expected fully charged settlement cost" must state how **failure within the budget** is treated. A method that quits cheaply without settling can have a lower mean spent cost than one that succeeds. Prespecify success probability by budget, cost conditional on success, cost of timeouts/failures or a common penalty, paired workloads/seeds, overhead charges, stopping rules, and statistical uncertainty. Compare the four configurations in §10 under equal **aggregate** resource accounting; report wall-clock performance separately.

Claim-by-claim proof audit

Manuscript item	ATLAS status	Reason
Prop. 2.5, path versus forced strategy	**PASS**	Existential single chess line does not establish the universal opponent-strategy claim; a checked derivation does establish derivability in a fixed calculus.
Def. 3.1 and Remark 3.2, bounded Win	**PASS with protocol qualification**	Sound distinction between actual bounded completion and object-theorem falsity; needs fixed trace and speaker semantics.
Def. 4.1, Player-0 activation	**Specification only**	Coherent partial rule, but legality must be furnished by an executable formal protocol.
Defs. 5.2–5.3, Eureka / IAE	**PASS as definitions**	Correctly distinguishes logically conservative proved lemmas from computational usefulness; no speedup theorem follows.
Lemma 6.2, no hypothesis laundering	**REPAIR REQUIRED**	Missing enforced consumer-side dependency/discharge rule.
Prop. 7.1, negation overlap	**PASS**	Derivability of $\neg c$ and $c \to \bot$ agrees under the stated negation convention; consistency/classicality may be relaxed.
Theorem 8.2, upward closure	**NOT ESTABLISHED AS WRITTEN**	May-simulate does not imply actual emission, and quotient membership can be ambiguous.
Theorem 8.4, minimal-antichain generation	**PASS**	Standard consequence of well-foundedness plus upper-closedness, with no finiteness conclusion.

Manuscript item	ATLAS status	Reason
Def. 8.6, task preorder	**CONDITIONAL PASS**	Reverse inclusion is reflexive/transitive whenever each settlement region is well-defined.
Prop. 8.7, negation symmetry	**CONDITIONAL PASS**	Holds for symmetric settlement with both certified double-negation directions admissible in the resource regime.
Prop. 9.1, optional-capability monotonicity	**PASS**	Old certified paths remain feasible if no compulsory tax or other change affects them.
Research Conjecture 9.2	**OPEN / NOT VERIFIED**	Empirical existence and cost advantage not established by the note.

A07 (Theorem 8.4 detail). In a well-founded partial order, every nonempty $U \cap {\downarrow}x$ has a minimal element, so $U = {\uparrow}\mathsf{Min}(U)$ for upper U; pairwise distinct minimal elements are incomparable. This does **not** mean there are finitely many generators: a countable antichain ordered only by equality is well-founded and has infinitely many minimal points. A finite-basis conclusion would require an additional assumption such as a well-quasi-order. Indeed global well-foundedness can be weakened to the local condition that every nonempty $U \cap {\downarrow}x$ has a minimal element.

A08 (Prop. 7.1 detail). If negation is defined as implication into bottom, $\neg c$ and $c \to \bot$ are identical or interderivable even in an intuitionistic setting; a consistent classical base is sufficient but not necessary. This is clarification, not a refutation.

A09 (roles and trust). The text thoughtfully preserves the historical 0/P/R/D/I/C roles and labels K/U as proposed variants. Retain this separation in every formal protocol and outcome table: *investigation agents*, *result labels*, and *truth statuses* are different objects. A countermodel or an independence claim may require a different checker or meta-level justification than a proof of c; tag the certificate type and trusted validator explicitly. This review has not checked the prose summaries of cited works [1]–[4] against the originals.

Required minimal repair set

1. **Choose a single settlement-region semantics** (existential attainable certificate or actual emitted certificate); change Definition 8.1, Theorem 8.2 and dependent results accordingly, including the equivalence-class invariant.
2. **Index resource-sensitive relations explicitly**; state admissible simulation cost maps and deadline behavior.
3. **Add the verified BANK import/discharge rule** to Lemma 6.2 and its proof.
4. **Specify speaker/parent/federation identity and handoff** in A's operational protocol, and identify A's actual search-policy change.
5. **Clarify SIC terminology** or show why mutual simulation equals the earlier intended SIC equivalence.
6. **Operationalize the experiment comparison**: outcome definitions, success accounting, costs for failures, baseline matching and logging.
7. **Keep status labels truthful**: conjectured speedup is untested; §8.4 is a valid conditional order-theoretic theorem; external source fidelity and independent mechanized checks remain unverified.

Final ATLAS judgment

Keep the manuscript and its thesis. The strongest part is the careful separation of a moving opponent, a fixed formal target, and a designated prover's future execution, together with the refusal to infer speedup from mere added capabilities. Its principal mathematical contribution is an elementary but potentially useful **conditional** upper-set/minimal-generator framework; that framework needs a repaired settledness semantics before it is sound as a single formal system. Nothing in this audit demonstrates basin escape, verified superiority of federation, or physical/empirical performance. No experiments, protocol changes, file replacements, or solver substitutions were performed.

Release recommendation: retain **WORKING NOTE / PROVISIONAL** until A01–A06 are addressed and then perform a second audit focused on exact definitions, proof obligations and source fidelity.